

SAMUEL LEVENTHAL

CONTACT INFORMATION

email sl@samleventhal.com
phone 1.801.808.6428
website www.samleventhal.com

ABOUT

Ph.D. in Computer Science from the University of Utah (Scientific Computing and Imaging Institute), with research spanning topological machine learning, graph neural networks, and scientific visualization. My work develops theoretically grounded ML methods — leveraging topological priors, discrete Morse theory, persistence filtration, and hierarchical GNNs — to address core challenges in graph learning (oversmoothing, heterophily) and scientific image analysis across materials science, medicine, and computational biology. Published work includes topologically informed GNN architectures for segmentation and representation learning (topological deep learning and segmentation, ICIP/NeurIPS), generative modeling via conditional GANs and scientific computing and visualization in HPC environments (PAVE framework, IEEE/ACM), and deep learning for medical diagnostics (lymphoma classification, American Journal of Clinical Pathology). Current work centers on topological neural operators that extend operator learning to simplicial and cellular complexes via copresheaf transports, and on learned persistence-filtration based methods for improving graph neural networks under heterophily. I aim to build scalable, interpretable ML frameworks integrating topology, geometry, and generative AI to advance discovery in scientific visualization, medical diagnostics, and computational physics.

EDUCATION

<i>Ph.D. Computer Science</i>	2024	University of Utah School of Computing & Scientific Computing and Imaging Institute
<i>B.Sc. Physics Minor: Philosophy</i>	2013	University of Utah Department of Physics and Astronomy & College of Humanities
<i>B.Sc. Mathematics Minor: Comp. Sci.</i>	2013	University of Utah Department of Mathematics & School of Computing

DISSERTATION COMMITTEE

Valerio Pascucci (Chair), Christopher R. Johnson, Dave Pugmire, Bei Wang Phillips, Aditya Bhaskara

EXPERIENCE

2025– <i>present</i>	Postdoctoral Research – Topological Neural Operators University of San Francisco Supervisor: Mustafa Hajj
2017–24	Ph.D. Research Scientist University of Utah, Scientific Computing & Imaging Institute – Ph.D. Mentor: Valerio Pascucci
2022	Visiting Research Scientist – GETO GNN: Topologically and Geometrically Constructed and Aware Graph Neural Networks Lawrence Livermore National Laboratory Mentors: Mark Heimann and Jayaraman J. Thiagarajan
2021	Teaching Mentorship – Programming for Engineers. Mentor: Valerio Pascucci
2020	Teaching Mentorship – Advanced Algorithms. Mentor: Aditya Bhaskara

- 2019 Visiting Research Scientist – PAVE: An In Situ Framework for Scientific Visualization and Machine Learning Coupling
Oak Ridge National Laboratory
Mentor: Mark Kim and David Pugmire
- 2018 Visiting Research Scientist – High dimensional space filling and sampling design with improved spectral sampling and applications Monte-Carlo Integration.
Lawrence Livermore National Laboratory
Mentor: Jayaraman J. Thiagarajan
- 2016 Independent Study – Topological Distortion as a Quality Measure for Dimension Reduction
University of Utah
Mentor: Bei Wang
- 2010-12 Independent Study – Search for Quantum like Structuring in Keplerian Systems.
University of Utah
Mentor: Stephan LeBohec

OTHER EXPERIENCE

- 2023 Conference Presentation
IEEE International Conference on Image Processing Graph Signal Processing and Machine Learning for Interpretable and robust image processing II
- 2019 Conference Presentation
Super Computing Conference 2019, 5th International Workshop on Data Analysis and Reduction for Big Scientific Data (DRBSD-5)

PAPERS IN PREPARATION

- 2026 Topological Neural Operators – Leventhal, Samuel, Lennart Bastian, Tolga Birdal, and Mustafa Hajj “Topological Neural Operators”. extended version in preparation for JMLR.
- 2026 Homophily-based Filtration Learning For Graph Neural Networks – Leventhal, Samuel, Attila Gyulassy, Mark Heimann, and Valerio Pascucci. In preparation.

PUBLICATIONS

PEER REVIEWED CONFERENCE PAPERS

- 2026 Topological Neural Operators – L. Bastian, S. Leventhal, M. Hajj, T. Birdal. “Topological Neural Operators”. NeurIPS, 2026 (under review) arXiv:2606.09806, 2026.
- 2023 Modeling Hierarchical Topological Structure in Scientific Images with Graph Neural Networks – Leventhal, Samuel, Attila Gyulassy, Valerio Pascucci, and Mark Heimann. “Modeling Hierarchical Topological Structure in Scientific Images with Graph Neural Networks.” International Conference on Image Processing. 2023.
- 2013 A Search for Quantum Like Structuring in Keplerian Systems – Samuel Leventhal and Stephan LeBohec. “A Search For Quantum Like Structuring in Keplerian Systems.” University of Utah Undergraduate Research Abstracts Journal, Vol. 13, Book 1. 2013

PEER REVIEWED JOURNAL PAPERS

- 2022 Exploring Classification of Topological Priors with Machine Learning for Feature Extraction – Leventhal, Samuel, Attila Gyulassy, Mark Heimann, and Valerio Pascucci. “Exploring Classification of Topological Priors with Machine Learning for Feature Extraction.” IEEE Transactions on Visualization and Computer Graphics (2023).
- 2020 Improving Augmented Human Intelligence to Distinguish Burkitt Lymphoma From Diffuse Large B-Cell Lymphoma Cases –Mohlman, Jeffrey S., Samuel D. Leventhal, Taft Hansen, Jessica Kohan, Valerio Pascucci, and Mohamed E. Salama.

"Improving augmented human intelligence to distinguish Burkitt lymphoma from diffuse large B-cell lymphoma cases." *American journal of clinical pathology* 153, no. 6 (2020): 743-759.

2019 High-throughput feature extraction for measuring attributes of deforming open-cell foams. – Petruzza, Steve, Attila Gyulassy, Samuel Leventhal, John J. Baglino, Michael Czabaj, Ashley D. Spear, and Valerio Pascucci. "High-throughput feature extraction for measuring attributes of deforming open-cell foams." *IEEE Transactions on Visualization and Computer Graphics* 26, no. 1 (2019): 140-150.

2018 Application of a Convolutional Neural Network to Distinguish Burkitt Lymphoma From Diffuse Large B-Cell Lymphoma – Mohlman, Jeffrey, Samuel Leventhal, Aniketh Venkat, Attila Gyulassy, Valerio Pascucci, and Mohamed Salama. "Application of a Convolutional Neural Network to Distinguish Burkitt Lymphoma From Diffuse Large B-Cell Lymphoma." *American Journal of Clinical Pathology* 150, no. suppl.1 (2018): Page S119-S119.

2011 Entropy Is Not Our Anomie – Leventhal, Samuel. *the Sponge*, Volume 2, Issue 1. p5-p11. 2011

PEER REVIEWED WORKSHOP PAPERS

2022 Modeling Hierarchical Topological Structure in Scientific Images with Graph Neural Networks – Leventhal, Samuel, Attila Gyulassy, Valerio Pascucci, and Mark Heimann. "Modeling Hierarchical Topological Structure in Scientific Images with Graph Neural Networks." In *NeurIPS 2022 Workshop: New Frontiers in Graph Learning*. 2022.

2019 PAVE: An In Situ Framework for Scientific Visualization and Machine Learning Coupling – Leventhal, Samuel, Mark Kim, and David Pugmire. "Pave: An in situ framework for scientific visualization and machine learning coupling." In *2019 IEEE/ACM 5th International Workshop on Data Analysis and Reduction for Big Scientific Data (DRBSD-5)*, pp. 8-15. IEEE, 2019.